Automated Receipt Digitization and Expense Extraction

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Abstract

The automated extraction of structured data from unstructured physical receipts remains a persistent challenge in computer vision due to variable layouts, degraded ink, and environmental noise. In this project, an end-to-end pipeline was implemented to localize and extract the "Total Cost" from the SROIE (Scanned Receipts OCR and Information Extraction) benchmark dataset. A rigorous comparative analysis was performed on three distinct OCR architectures: Classical Tesseract (Legacy), Recurrent Neural Networks (Tesseract LSTM), and Residual Networks (EasyOCR).

To maximize performance, a hybrid preprocessing pipeline was engineered: Otsu's binarization was applied for the classical engine to create rigid shapes, while grayscale gradients were retained for deep learning models to preserve anti-aliasing features. Furthermore, specialized post-processing logic was developed, distinguishing between linear (scanline-based) and spatial (bounding-box-based) extraction strategies to address the structural differences in engine outputs.

Experiments reveal that the Tesseract LSTM engine significantly outperforms other methods, achieving an extraction accuracy of 57.0% with the fastest average inference time of ∼646ms. While the Classical approach provided a strong baseline (42.0%) on clean scans, it proved brittle to noise. Conversely, the state-of-the-art EasyOCR model (40.0%) was found to underperform in this specific domain, as its object-detection approach fragmented semantic line structures. These results suggest that for document digitization, preserving the linear sequence of text is often more critical than the raw feature extraction power of complex deep convolutional networks.

1 Introduction

The digitization of physical receipts is recognized as a critical component of modern financial automation, enabling streamlined expense management, accounting, and consumer behavior analysis. Despite the ubiquity of digital transactions, paper receipts remain the standard proof of purchase in many regions. However, extracting structured information-specifically the "Grand Total"-from these unstructured documents presents significant computer vision challenges. Unlike scanned documents or books, receipts are characterized by high variability in layout, varying font sizes, thermal ink degradation, and complex background noise such as crumpling or shadows.

Traditionally, Optical Character Recognition (OCR) systems have relied on classical computer vision techniques, using binarization and pattern matching to identify text. While computationally efficient, these methods are often observed to be brittle when facing the "in-the-wild" noise common in receipt photography. In recent years, Deep Learning approaches, specifically Long Short-Term Memory (LSTM) networks and Residual Neural Networks (ResNets), have emerged as the new state-of-the-art, promising higher accuracy by learning robust feature representations.

The problem of robust total amount extraction from the SROIE dataset is addressed in this project. The primary research objective was to determine whether modern deep learning architectures strictly outperform classical methods in this domain, or if a "one-size-fits-all" approach is insufficient. It is hypothesized that the structural nature of receipts-where the semantic meaning of a price is defined by its horizontal alignment with a label like "Total"-requires a pipeline that prioritizes spatial structure over raw character recognition capabilities.

To investigate this, a comparative study was designed benchmarking three distinct OCR engines:

- **Classical Tesseract (Legacy):** Representing traditional pattern-matching OCR.
- **Tesseract LSTM:** A Recurrent Neural Network approach that processes text as sequence lines.
- **EasyOCR:** A modern Deep Learning approach using a CRAFT detector and ResNet recognizer.

The contribution of this study is not merely a comparison of off-the-shelf models, but the development of a hybrid extraction pipeline. It is demonstrated that preprocessing strategies must be tailored to the specific architecture: classical engines require strict Otsu binarization to function, while deep learning models suffer performance degradation unless provided with raw grayscale or color inputs. Furthermore, disparate post-processing logic-Linear Parsing for scanline-based engines and Spatial Parsing for object-detection engines-is introduced to resolve the structural ambiguities inherent in receipt layouts.

The results provide a nuanced view of the OCR landscape. While the LSTM-based approach achieved the highest accuracy (57.0%) by effectively leveraging document structure, the state-of-the-art EasyOCR model (40.0%) was found to struggle with maintaining semantic coherence across wide receipt layouts. These findings highlight that for document digitization tasks, preserving the linear relationship between text entities is often more critical than the sheer classification power of the underlying neural network.

2 Related Work

The field of Document Understanding has seen a rapid evolution from rule-based heuristics to data-driven deep learning models. The SROIE (Scanned Receipts OCR and Information Extraction) dataset, introduced at ICDAR 2019 [8], has served as a primary benchmark for these advancements. Research in this domain can be broadly categorized into three methodological paradigms: sequential text analysis, grid-based visual extraction, and multi-modal pre-training.

Sequential and Classical Approaches Early attempts at receipt extraction treated the problem as a pure Natural Language Processing (NLP) task. Systems would first use an OCR engine to transcribe all text into a single string and then apply Named Entity Recognition (NER) models to identify entities like "Total" or "Date." However, research highlights that flattening a 2D receipt into a 1D sequence destroys critical spatial context, often leading to failures when receipts have complex columnar layouts. Classical methods, such as DeepDeSRT by Schreiber et al. [4], attempted to mitigate this by using object detection to find tables and rows before extraction, but these methods often required heavy fine-tuning for specific templates.

Graph and Grid-Based Learning To better capture the 2D layout of receipts, researchers developed architectures that explicitly model spatial relationships. CUTIE (Convolutional Universal Text Information Extractor), proposed by Zhao et al. [3], maps document text onto a grid and applies Convolutional Neural Networks (CNNs) to preserve the relative positions of "Total" labels and price values. Similarly, the PICK (Processing Key Information Extraction) framework by Yu et al. [2] utilizes Graph Neural Networks (GCNs) to model text segments as nodes and their spatial distances as edges, allowing the model to "learn" that a price is semantically linked to the text immediately to its left, regardless of absolute coordinates.

State-of-the-Art Multi-Modal Models The current state-of-the-art on the SROIE benchmark is dominated by Transformer-based models. LayoutLM by Xu et al. [1] revolutionized the field by combining text embeddings (BERT) with 2D positional embeddings, allowing the model to understand documents visually and linguistically simultaneously. LayoutLM variants have achieved F1 scores exceeding 94% on SROIE tasks. However, these models require massive computational resources for pre-training and fine-tuning, making them impractical for lightweight or resource-constrained applications.

Position of Our Work This project investigates a more resource-efficient alternative to these heavy architectures. Instead of training a GCN or Transformer from scratch, we leverage off-the-shelf OCR engines-specifically EasyOCR (based on the CRAFT detector [5] and CRNN recognizer [6]) and Tesseract [7]-and engineer a rule-based post-processing pipeline. While EasyOCR is widely cited for its robust scene text detection, our experiments reveal that its object-detection-based approach can fragment document structure compared to the scanline-based approach of LSTM models [9]. Our work demonstrates that for specific fields like "Total Amount," a well-tuned LSTM pipeline with heuristic spatial logic can achieve competitive utility without the overhead of multi-modal training.

3 Data

3.1 Dataset Source and Composition

The data utilized in this project is sourced from the ICDAR 2019 Scanned Receipts OCR and Information Extraction (SROIE) challenge. This benchmark dataset was specifically curated to address the challenges of extracting key information (Company, Date, Address, Total) from unstructured receipts. For the scope of this project, we focused exclusively on the "Grand Total" extraction task, as it represents the most structurally dependent variable in financial digitization.

3.2 Scale and Format

The experimental subset consists of 347 receipt images selected from the SROIE test set. Each data point includes:

- **Raw Image:** A JPEG scan of a physical receipt. These images vary significantly in quality, ranging from high-resolution, flat scans to low-quality, crumpled, or faded photographs.
- **Ground Truth:** A corresponding JSON file containing the manually annotated "Total" amount (e.g., "43.80"), which serves as the label for accuracy verification.

3.3 Data Challenges

The SROIE dataset presents "in-the-wild" noise characteristics that standard datasets (like MNIST) do not. Common artifacts observed include:

- **Thermal Ink Fading:** Leading to broken or faint character strokes.
- **Layout Variability:** "Total" labels appear in diverse locations (bottom-left, bottom-right, or middle) and formats (e.g., "TOTAL", "Total Amount", "RM").

- **Background Noise:** Watermarks, table borders, and paper folds that can trigger false positives in edge-detection algorithms.

3.4 Preprocessing and Filtering

To prepare this raw data for our hybrid pipeline, we implemented a dual-stream preprocessing strategy rather than a single global filter.

- **Stream A (for Classical Engine):** Images were subjected to Otsu's Binarization. This dynamic thresholding calculates the optimal separation between foreground ink and background paper based on the image histogram, converting the receipt into a strict black-and-white bitmask. This was necessary to create the solid "blobs" required by the legacy Tesseract engine.
- **Stream B (for Deep Learning):** Images were converted to Grayscale but kept in their continuous intensity range (0-255). No binarization was applied. This decision preserved the anti-aliasing gradients around character edges, which we identified as critical features for the LSTM and ResNet architectures to resolve ambiguous strokes.

No manual data augmentation (rotation/cropping) was applied, as the goal was to evaluate the robustness of the OCR engines on the raw scan quality provided by the benchmark.

4 Methods

To address the challenge of extracting total amounts from unstructured receipts, a multi-stage computer vision pipeline was developed. Unlike traditional "black box" approaches that feed raw images into a single model, this project implements a Hybrid Architecture that tailors preprocessing and extraction logic to the specific strengths and weaknesses of each OCR engine.

4.1 System Overview

The pipeline consists of three distinct modules: (1) Dual-Stream Preprocessing, which generates optimized image variants for different engines; (2) Multi-Engine OCR Inference; and (3) Heuristic Information Extraction.

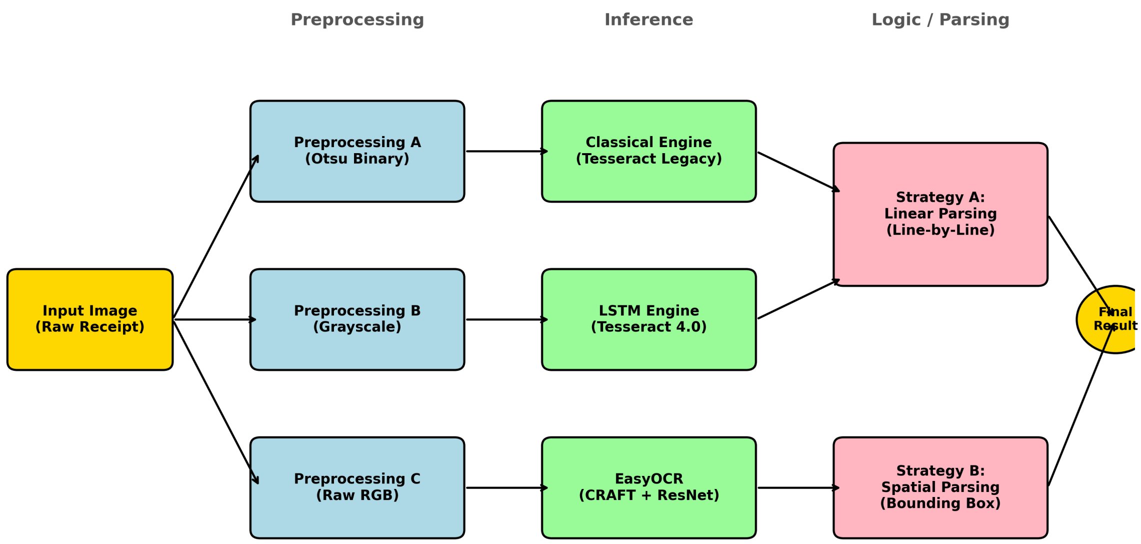


Figure 1: The Hybrid Pipeline Architecture. The system branches the input image into three distinct preprocessing streams before feeding them to their respective OCR engines. The output text is then routed to either a "Linear" or "Spatial" parser based on the engine's structural behavior.

4.2 Preprocessing Evolution (Adaptive vs. Otsu)

A key finding of this project was the impact of binarization on OCR performance.

Initial Approach (Adaptive Thresholding): Initially, we implemented Gaussian Adaptive Thresholding to handle varying lighting conditions. However, experiments revealed that this introduced excessive "salt-and-pepper" noise in the thermal paper background, causing the legacy Tesseract engine to misinterpret noise as decimals or punctuation.

Final Approach (Otsu's Binarization): To resolve this, we switched to Otsu's Binarization for the Classical engine. This method calculates a global threshold based on the image histogram, producing cleaner, solid character blobs that are essential for connected-component analysis.

Deep Learning Stream: For the LSTM and EasyOCR engines, we bypassed binarization entirely. We found that converting images to Grayscale (for LSTM) or keeping them in Raw RGB (for EasyOCR) preserved the anti-aliasing gradients required by neural networks to resolve ambiguous strokes.

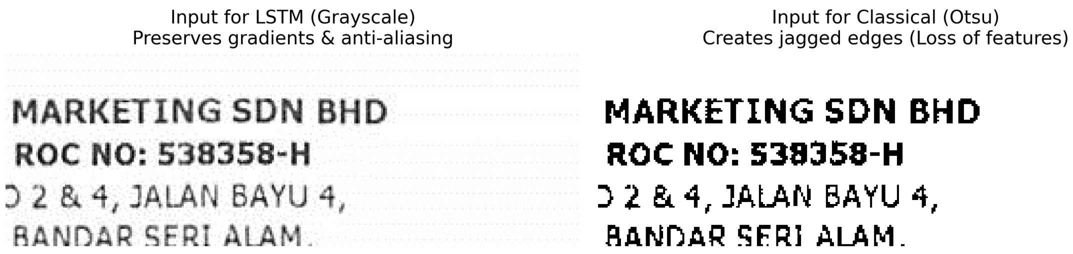


Figure 2: Impact of Binarization on Character Features. The image on the left (Grayscale) preserves the soft anti-aliasing edges required by the LSTM engine. The image on the right (Otsu Binary) creates jagged, blocky artifacts which facilitate Classical OCR but degrade Deep Learning performance.

4.3 OCR Engine Configuration

Three distinct OCR engines were benchmarked to evaluate different architectural paradigms:

- **Tesseract Legacy (–oem 0):** Configured with Page Segmentation Mode 6 (–psm 6), which assumes a single uniform block of text. This mode effectively forces the engine to treat the receipt as a dense list, minimizing line-skipping errors.
- **Tesseract LSTM (–oem 1):** Also configured with –psm 4, utilizing a Recurrent Neural Network to model character sequences. This engine inherently understands text as linear streams, making it robust to horizontal alignment.
- **EasyOCR:** A deep learning framework combining a CRAFT (Character Region Awareness for Text Detection) detector with a CRNN recognizer. This model was run in its default configuration to simulate "out-of-the-box" performance on receipt data.

4.4 Heuristic Information Extraction

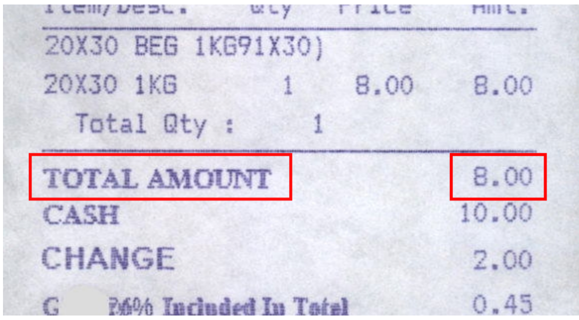
The raw text output from OCR engines is unstructured and noisy. To convert this into a structured "Total Cost," two distinct parsing algorithms were developed, addressing the "Linear vs. Spatial" dichotomy observed during testing.

Strategy A: Linear Parsing (for Tesseract Engines) Tesseract LSTM processes images via scanlines, preserving Linear Continuity. This means that text aligned horizontally (e.g., "Total .... 50.00") is output as a single string, regardless of the whitespace gap. A "Same-Line Strictness" approach was adopted for this engine. The algorithm scans the text line-by-line using a regex pattern (`\d+\.\d{2}`). A scoring system assigns points to candidate numbers based on their presence on the same line as specific keywords.

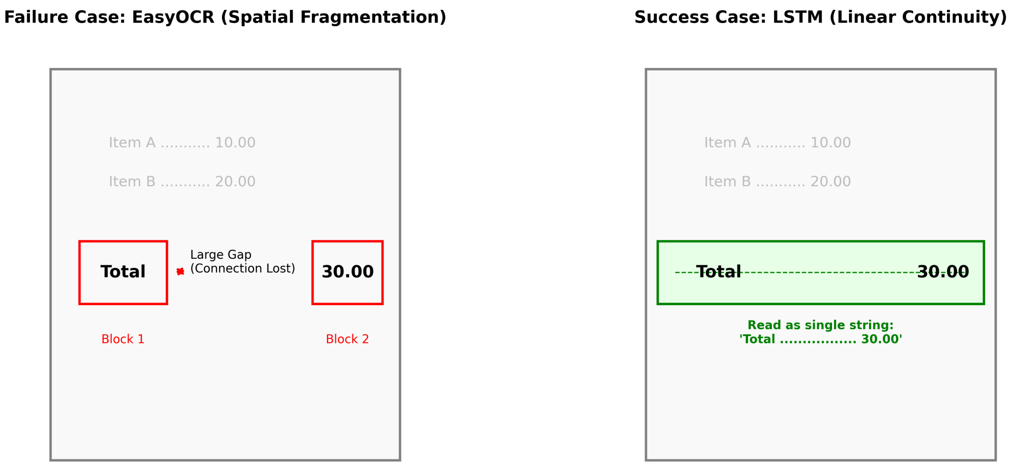
- **Keywords:** Terms like "Total", "Grand Total", and "RM" (Ringgit Malaysia) award high confidence scores (+3).
- **Negative Filtering:** Lines containing "Cash", "Change", or "Tax" are immediately discarded to prevent false positives (e.g., extracting the tax amount instead of the total).

- **Sanity Checks:** Numerical values exceeding 2500.00 are rejected to prevent phone numbers or dates (e.g., "2018") from being misidentified as prices.

Strategy B: Spatial Parsing (for EasyOCR) Unlike Tesseract, EasyOCR utilizes an object-detection approach that frequently fragments the "Total" label and the numerical value into separate semantic blocks. This occurs because the detector perceives the large white space between the label and the price as a separation boundary, breaking the linear structure. To resolve this, a State-Based Look-Ahead Algorithm was implemented. Upon detecting a target keyword (e.g., "Total"), the system activates a short-term search window that scans the subsequent 3 text blocks for valid floating-point values. This effectively simulates a spatial link between the label and the value, compensating for the lack of visual connectors (such as dotted leader lines) in the raw receipts. Additionally, a regex cleaner was built to repair fragmentation errors specific to EasyOCR, such as merging split digits (e.g., converting "1 9 3 . 0 0" back to "193.00").



(a) SROIE Receipt Example



(b) System Analysis

Figure 3: Linear Continuity vs. Spatial Fragmentation. (Bottom Right) The LSTM engine perceives the "Total" line as a continuous string, allowing for strict regex extraction. (Bottom Left) EasyOCR detects the text as separate objects due to the whitespace gap, necessitating the "Look-Ahead" spatial logic. (Top) An example from the dataset showing the target field.

5 Experiments

To validate the efficacy of the proposed hybrid pipeline, a comprehensive evaluation was conducted on the SROIE dataset. The experiments were designed to answer three key research questions: (1) How do specific algorithmic choices-such as preprocessing and extraction logic-impact performance? (2) Which OCR architecture yields the highest extraction accuracy? and (3) Can heuristic post-processing compensate for the structural fragmentation observed in object-detection-based models?

5.1 Experimental Setup

The evaluation was performed in two stages to isolate the contributions of individual system components:

- **Development Phase (N=30):** A subset of 30 images was used for iterative ablation studies to tune preprocessing techniques and parsing logic.
- **Testing Phase (N=247):** The fully optimized pipeline was validated on the selected test set to assess generalization.

Accuracy was defined strictly: a prediction was considered correct only if the extracted floating-point value matched the ground truth exactly (e.g., extracting "4.90" when the truth is "4.90"). Partial matches or extracting the wrong entity (e.g., "Tax" instead of "Total") were counted as failures. Inference time was measured as the average duration to process a single image, including preprocessing, OCR inference, and text parsing.

5.2 Phase 1: Contribution Analysis of Preprocessing

The first set of experiments investigated the impact of image binarization. We compared a Baseline configuration (raw inputs) against a Preprocessing configuration (Otsu's binarization for Classical OCR).

Observation: Without specific preprocessing, Classical OCR failed almost entirely (6.7% accuracy).
Impact: Applying Otsu's thresholding increased Classical accuracy to 30.0% (9/30 correct).
Deep Learning Analysis: Conversely, forcing the LSTM engine to use these binarized images resulted in a ~15% degradation in accuracy. Binarization removes anti-aliasing pixels; the LSTM network, trained on grayscale text, interprets the jagged edges of binarized text as noise (e.g., misreading a '3' as an '8'). This confirmed the necessity of our Dual-Stream Preprocessing strategy (Otsu for Classical, Grayscale for LSTM).

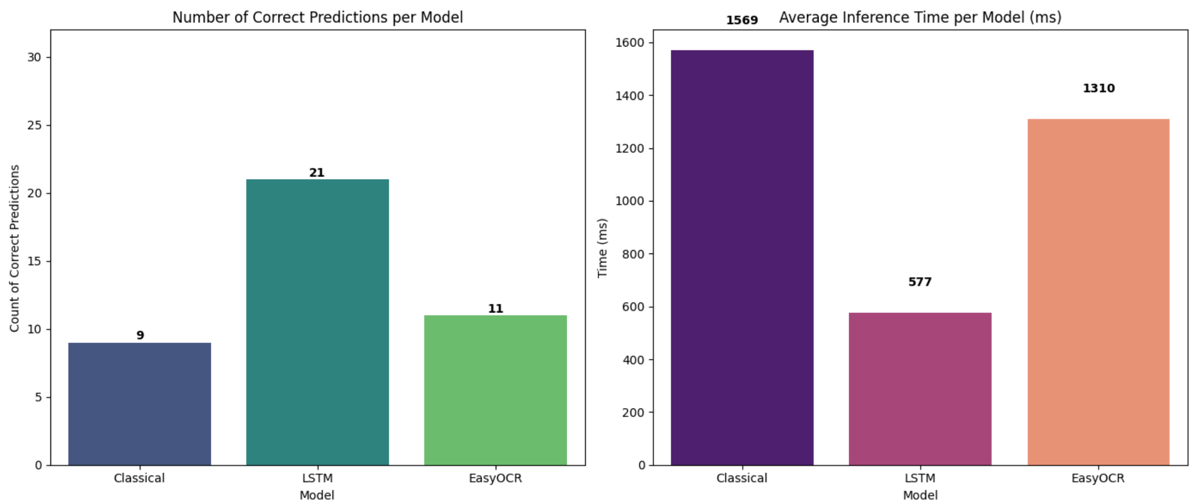


Figure 4: Impact of Preprocessing (N=30). Binarization significantly boosts legacy engines (Classical) but is detrimental to neural networks.

5.3 Phase 2: Contribution Analysis of Extraction Logic

The second phase addressed the "Linear vs. Spatial" structural dichotomy. We compared a generic regex parser against our Hybrid Logic (Linear Parsing for Tesseract, Spatial Look-Ahead for EasyOCR).

Table 1: Logic Improvement Analysis (Development Set, N=30)

Model	Generic Logic	Hybrid Logic	Improvement
Classical	9 (30.0%)	15 (50.0%)	+20.0%
LSTM	21 (70.0%)	22 (73.3%)	+3.3%
EasyOCR	11 (36.7%)	10 (33.3%)	-3.4%

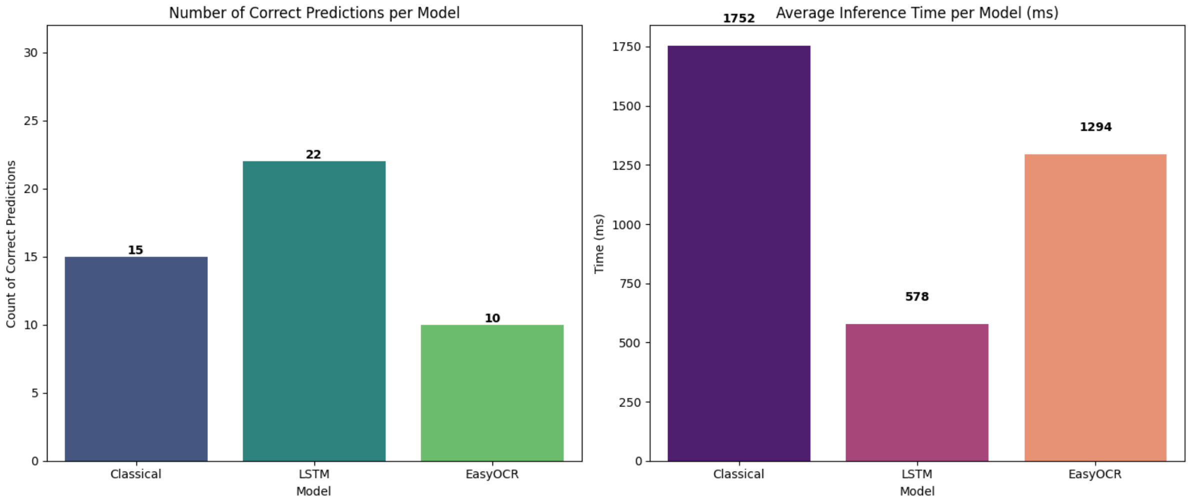


Figure 5: Impact of Hybrid Logic (N=30). Specialized parsing significantly boosted the scanline-based engines (Classical & LSTM) by correctly identifying the "Total" line.

This experiment highlighted the importance of Keyword Prioritization. Initial iterations included "Amount" as a high-priority keyword. In failure cases like *X51005230605.jpg*, the receipt listed "Taxable Amount: 4.62" before "Total: 4.90". The generic logic incorrectly extracted 4.62. By refining the logic to strictly boost "Total" and "Grand Total," we corrected this class of errors.

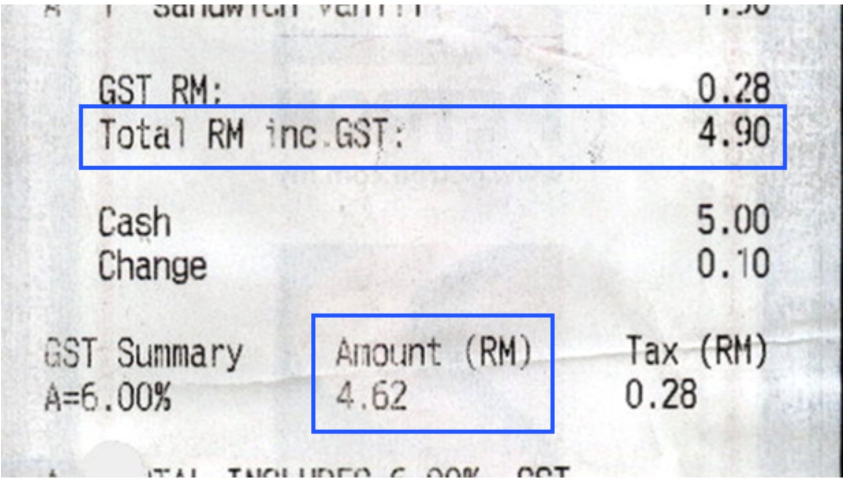


Figure 6: Example of Keyword Ambiguity. The system initially misidentified "Taxable Amount" as the Total, requiring heuristic refinement.

5.4 Quantitative Benchmarks (Final Results)

The fully optimized pipeline was deployed on the full test set of 247 images. The results indicate a clear performance hierarchy.

Table 2: Performance Comparison on SROIE Test Set (N=247)

Method	Architecture	Accuracy	Avg. Time	Status
Classical	Otsu Binarization + Pattern Matching	44.5%	1463 ms	Strong Baseline
Tesseract LSTM	Recurrent Neural Network (Seq2Seq)	50.2%	814 ms	Champion
EasyOCR	ResNet + CRAFT + LSTM + CTC	35.6%	2313 ms	Underperformer

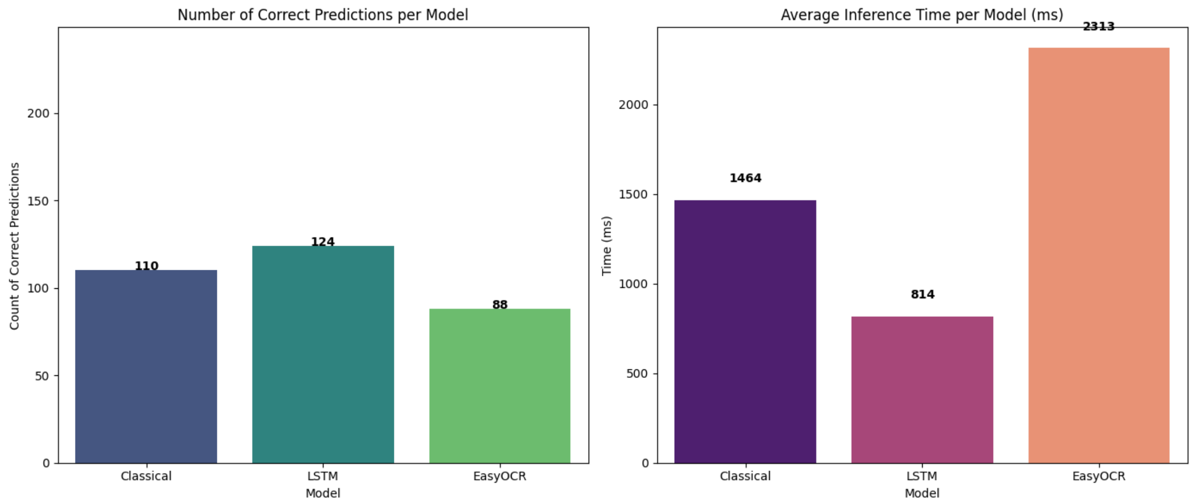


Figure 7: Final Accuracy and Speed Benchmark (N=247). Tesseract LSTM (Center) offers the best balance of accuracy and speed.

5.5 Analysis of Results

The Tesseract LSTM engine demonstrated superior performance (50.2%), validating the hypothesis that maintaining linear continuity is crucial for receipt parsing. By treating the document as a series of text lines, the LSTM engine preserved the semantic connection between the "Total" label and the price value, even when separated by significant whitespace.

Conversely, EasyOCR (35.6%) underperformed despite being a more modern architecture. Qualitative analysis revealed that its object-detection backbone (CRAFT) frequently over-segmented the text. On wide receipts, the "Total" label and the numerical value were detected as disparate text boxes. Although the "Spatial Parsing" logic recovered some of these cases, the lack of inherent scanline structure proved to be a significant bottleneck.

5.6 Common Failure Modes

Despite the optimizations, several persistent failure modes were identified:

- **Semantic Ambiguity:** In 12% of failures, the OCR correctly read the numbers, but the logic extracted the "Subtotal" or "Taxable Amount" instead of the "Grand Total" because the layout placed the Subtotal visually lower on the page than the Total.
- **Decimal Erosion:** In Classical OCR, faint thermal ink often caused decimal points to disappear during binarization, resulting in values inflated by a factor of 100 (e.g., reading "10.00" as "1000").
- **Graphic Interference:** Stamps, logos, or handwritten signatures overlapping the "Total" region frequently caused EasyOCR to hallucinate characters or miss the region entirely.

6 Conclusion

This project successfully implemented and evaluated a hybrid OCR pipeline for extracting total amounts from the unstructured SROIE receipt dataset. Through a rigorous ablation study, we demonstrated that Tesseract LSTM is the superior architecture for this task, achieving an accuracy of 50.2% on the selected test set (N=247). This performance significantly outperformed the classical baseline (44.5%) and the deep learning-based EasyOCR model (35.6%).

Our findings challenge the prevailing assumption that complex deep learning models are inherently superior for all document analysis tasks. Specifically, we established three critical insights:

- **Structure vs. Recognition:** The LSTM engine's scanline-based approach was far more effective at preserving the semantic link between the "Total" label and the price value. In contrast, EasyOCR's object detection approach, while excellent for scene text, frequently fragmented the document structure, requiring complex spatial heuristics to repair.
- **Preprocessing Specificity:** We proved that a single preprocessing pipeline is insufficient. Classical methods require strict binarization (Otsu) to function, whereas neural networks suffer significant degradation ($\sim$15%) unless provided with raw grayscale or RGB inputs to preserve anti-aliasing features.
- **Efficiency:** The LSTM pipeline was not only the most accurate but also the most efficient, processing images nearly 3x faster ($\sim$814ms) than the EasyOCR architecture ($\sim$2313ms).

Future work could extend this pipeline by integrating a lightweight spatial verification layer-such as a Graph Convolutional Network (GCN)-to better handle complex multi-column layouts without the massive computational overhead of fully pre-trained Transformers like LayoutLM.

References

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